

Data Mining

Chapter 4. Data Warehousing and On-line Analytical Processing

Esha Nawaz

Lecturer

The University of Faisalabad

- ❑ Data Warehouse: Basic Concepts
- ❑ Data Warehouse Modeling: Data Cube and OLAP
- ❑ Data Warehouse Design and Usage
- ❑ Data Warehouse Implementation
- ❑ Summary

- ❑ Defined in many different ways, but not rigorously
 - ❑ A decision support database that is maintained separately from the organization's operational database
 - ❑ Support information processing by providing a solid platform of consolidated, historical data for analysis
- ❑ “A data warehouse is a subject-oriented, integrated, time-variant, and nonvolatile collection of data in support of management's decision-making process.” —W. H. Inmon
- ❑ Data warehousing:
 - ❑ The process of constructing and using data warehouses

- ❑ Organized around major subjects, such as customer, product, sales
- ❑ Focusing on the modeling and analysis of data for decision makers, not on daily operations or transaction processing
- ❑ Provide a simple and concise view around particular subject issues by excluding data that are not useful in the decision support process

- ❑ Constructed by integrating multiple, heterogeneous data sources
 - ❑ relational databases, flat files, on-line transaction records
- ❑ Data cleaning and data integration techniques are applied.
 - ❑ Ensure consistency in naming conventions, encoding structures, attribute measures, etc. among different data sources
 - ❑ Ex. Hotel price: differences on currency, tax, breakfast covered, and parking
 - ❑ When data is moved to the warehouse, it is converted

- ❑ The time horizon for the data warehouse is significantly longer than that of operational systems
 - ❑ Operational database: current value data
 - ❑ Data warehouse data: provide information from a historical perspective (e.g., past 5-10 years)
- ❑ Every key structure in the data warehouse
 - ❑ Contains an element of time, explicitly or implicitly
 - ❑ But the key of operational data may or may not contain “time element”

- ❑ Independence
 - ❑ A physically separate store of data transformed from the operational environment
- ❑ Static: Operational update of data does not occur in the data warehouse environment
 - ❑ Does not require transaction processing, recovery, and concurrency control mechanisms
 - ❑ Requires only two operations in data accessing:
 - ❑ *initial loading of data and access of data*

OLTP vs. OLAP

❑ OLTP: Online transactional processing

❑ DBMS operations

❑ Query and transactional processing

❑ OLAP: Online analytical processing

❑ Data warehouse operations

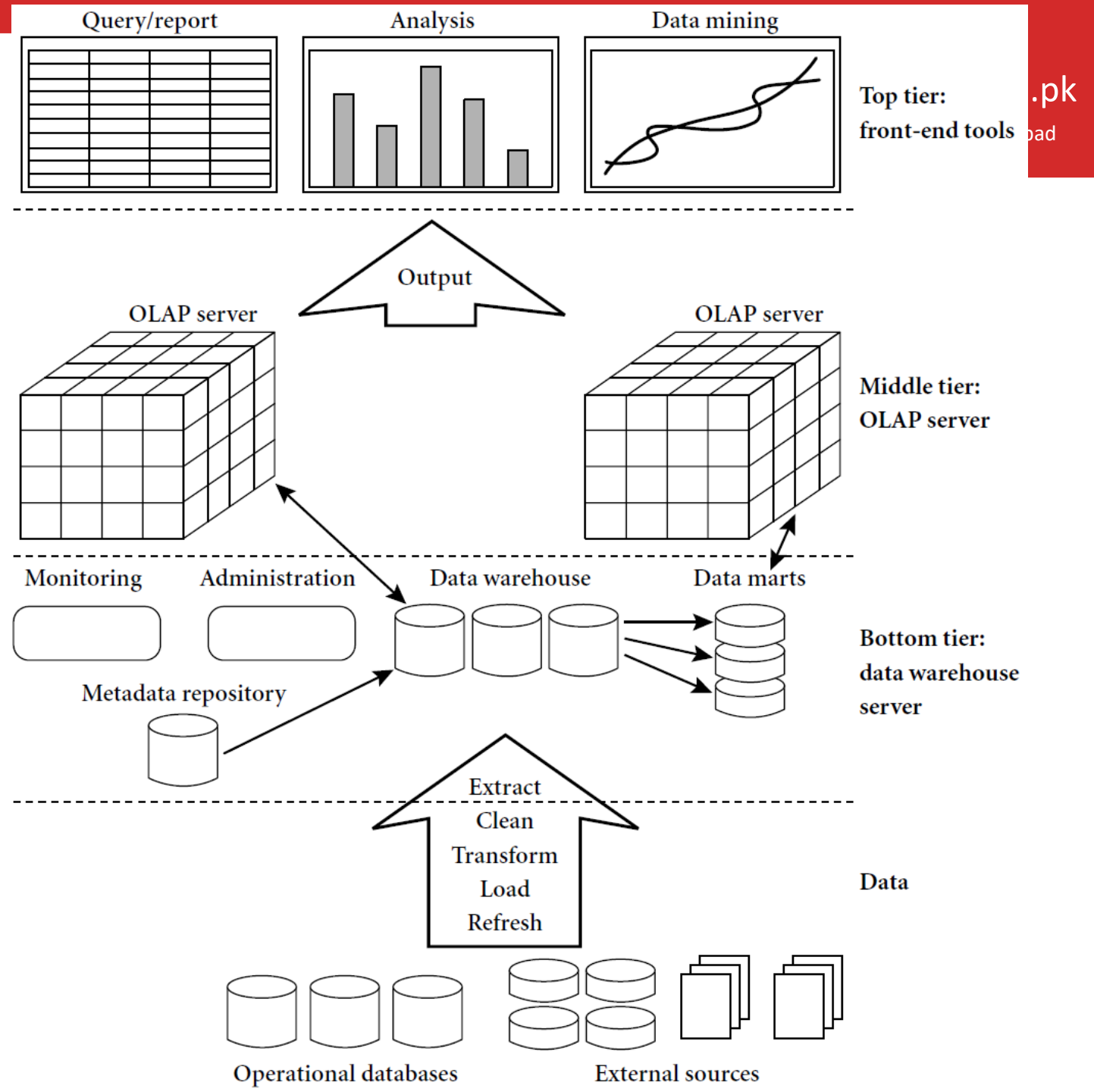
❑ Drilling, slicing, dicing, etc.

	OLTP	OLAP
users	clerk, IT professional	knowledge worker
function	day to day operations	decision support
DB design	application-oriented	subject-oriented
data	current, up-to-date detailed, flat relational isolated	historical, summarized, multidimensional integrated, consolidated
usage	repetitive	ad-hoc
access	read/write index/hash on prim. key	lots of scans
unit of work	short, simple transaction	complex query
# records accessed	tens	millions
#users	thousands	hundreds
DB size	100MB-GB	100GB-TB
metric	transaction throughput	query throughput, response

- ❑ High performance for both systems
 - ❑ DBMS— tuned for OLTP: access methods, indexing, concurrency control, recovery
 - ❑ Warehouse—tuned for OLAP: complex OLAP queries, multidimensional view, consolidation
- ❑ Different functions and different data:
 - ❑ missing data: Decision support requires historical data which operational DBs do not typically maintain
 - ❑ data consolidation: DS requires consolidation (aggregation, summarization) of data from heterogeneous sources
 - ❑ data quality: different sources typically use inconsistent data representations, codes and formats which have to be reconciled
- ❑ Note: There are more and more systems which perform OLAP analysis directly on relational databases

Data Warehouse: A Multi-Tiered Architecture

- ❑ Top Tier: Front-End Tools
- ❑ Middle Tier: OLAP Server
- ❑ Bottom Tier: Data Warehouse Server
- ❑ Data



❑ Enterprise warehouse

- ❑ Collects all of the information about subjects spanning the entire organization

❑ Data Mart

- ❑ A subset of corporate-wide data that is of value to a specific groups of users
- ❑ Its scope is confined to specific, selected groups, such as marketing data mart
 - ❑ Independent vs. dependent (directly from warehouse) data mart

❑ Virtual warehouse

- ❑ A set of views over operational databases
- ❑ Only some of the possible summary views may be materialized

❑ Data extraction

- ❑ get data from multiple, heterogeneous, and external sources

❑ Data cleaning

- ❑ detect errors in the data and rectify them when possible

❑ Data transformation

- ❑ convert data from legacy or host format to warehouse format

❑ Load

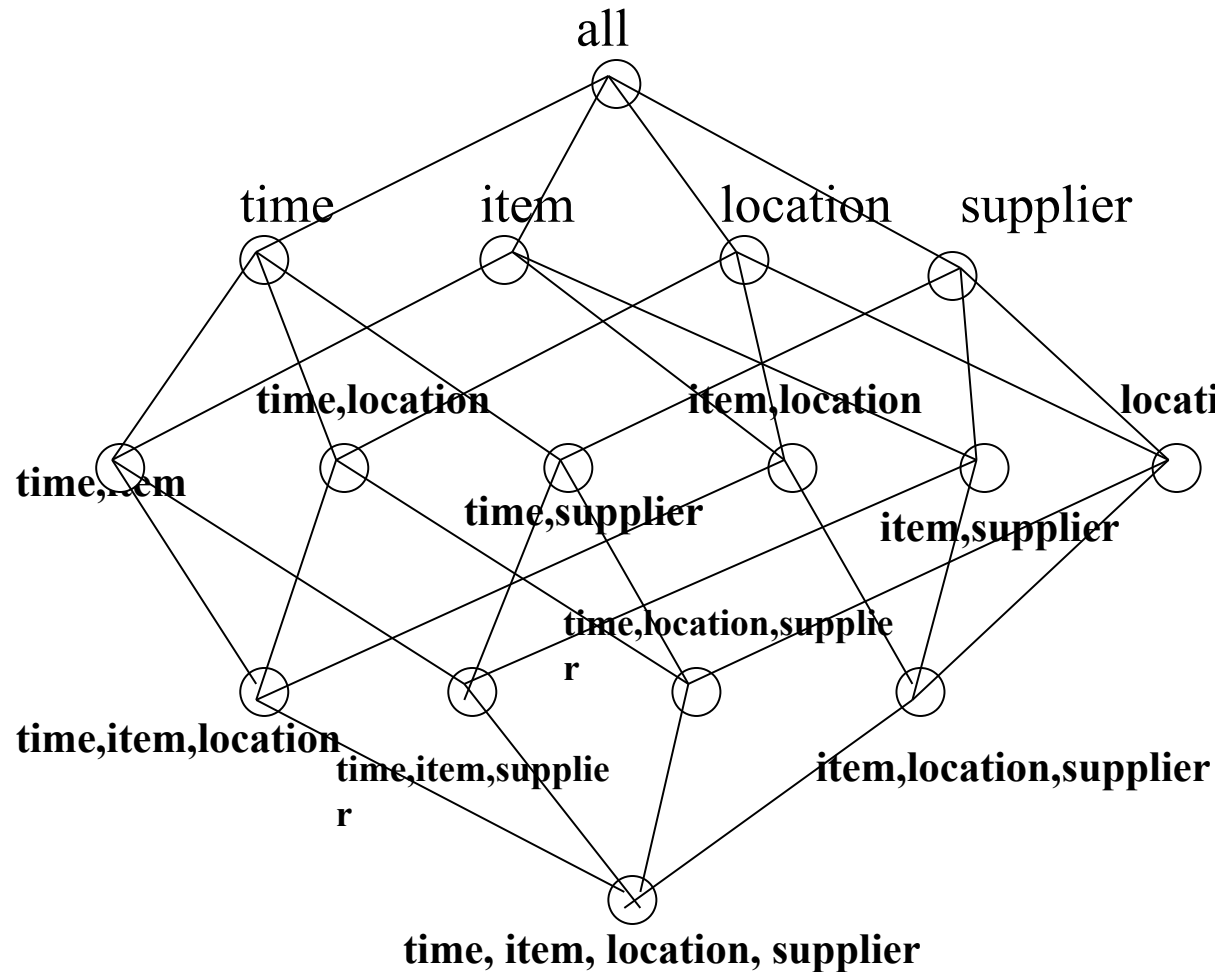
- ❑ sort, summarize, consolidate, compute views, check integrity, and build indices and partitions

❑ Refresh

- ❑ propagate the updates from the data sources to the warehouse

- ❑ **Meta data** is the data defining warehouse objects. It stores:
 - ❑ Description of the structure of the data warehouse
 - ❑ schema, view, dimensions, hierarchies, derived data defn, data mart locations and contents
 - ❑ Operational meta-data
 - ❑ data lineage (history of migrated data and transformation path), currency of data (active, archived, or purged), monitoring information (warehouse usage statistics, error reports, audit trails)
 - ❑ The algorithms used for summarization
 - ❑ The mapping from operational environment to the data warehouse
 - ❑ Data related to system performance
 - ❑ warehouse schema, view and derived data definitions
 - ❑ Business data
 - ❑ business terms and definitions, ownership of data, charging policies

- ❑ A **data warehouse** is based on a multidimensional data model which views data in the form of a data cube
- ❑ A data cube, such as sales, allows data to be modeled and viewed in multiple dimensions
 - ❑ **Dimension tables**, such as item (item_name, brand, type), or time(day, week, month, quarter, year)
 - ❑ **Fact table** contains **measures** (such as dollars_sold) and keys to each of the related dimension tables
- ❑ **Data cube**: A lattice of cuboids
 - ❑ In data warehousing literature, an n-D base cube is called a **base cuboid**
 - ❑ The top most 0-D cuboid, which holds the highest-level of summarization, is called the **apex cuboid**
 - ❑ The lattice of cuboids forms a **data cube**.



0-D (*apex*)
cuboid

1-D cuboids

2-D cuboids

3-D cuboids

4-D (*base*)
cuboid

❑ Key Features of Fact Table

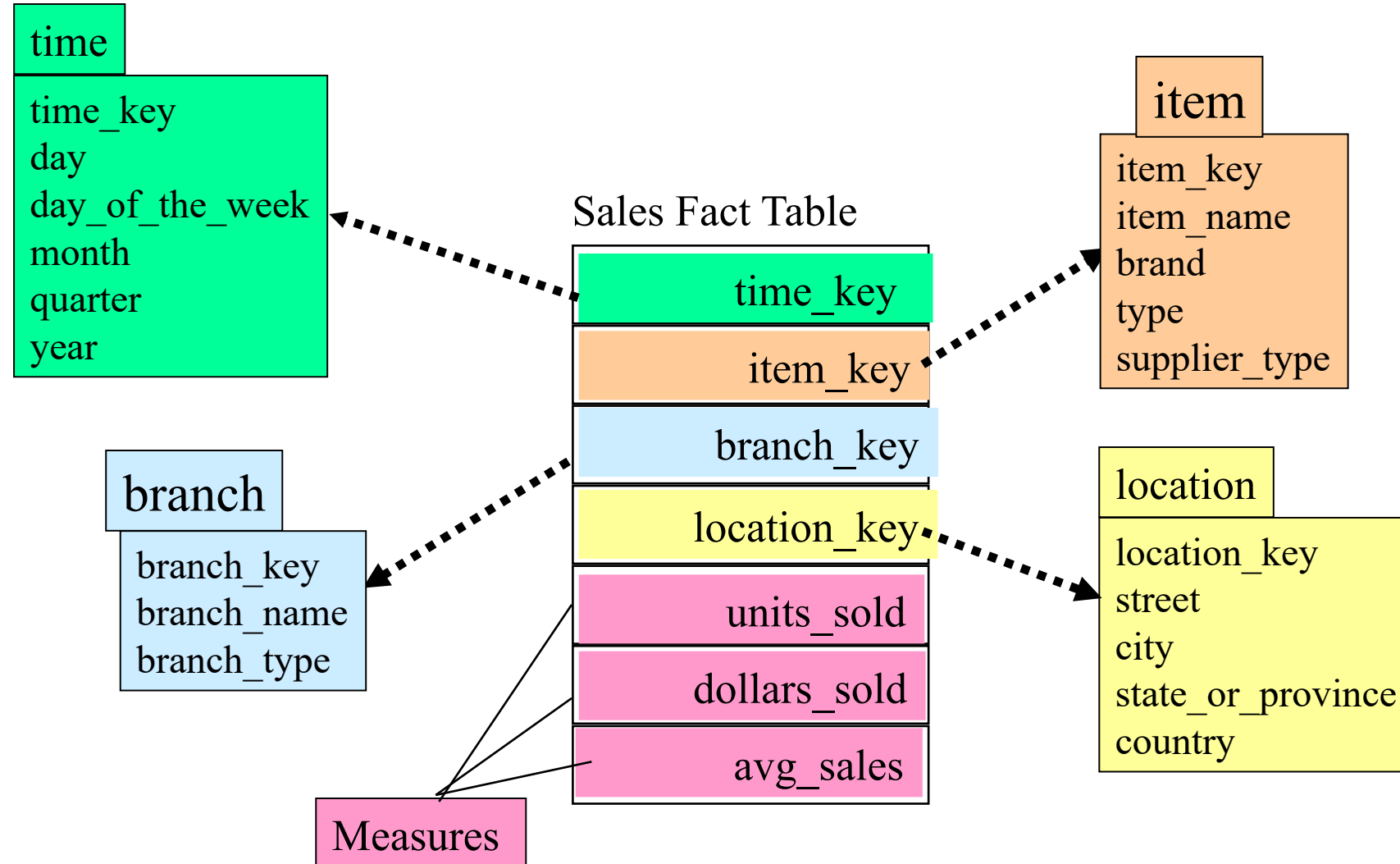
- ❑ Contains numbers (facts) like sales, profit, quantity
- ❑ Very large in size (millions of rows)
- ❑ Has foreign keys to other tables (like product, customer, date)
- ❑ Used for analysis and reporting
- ❑ Grows continuously

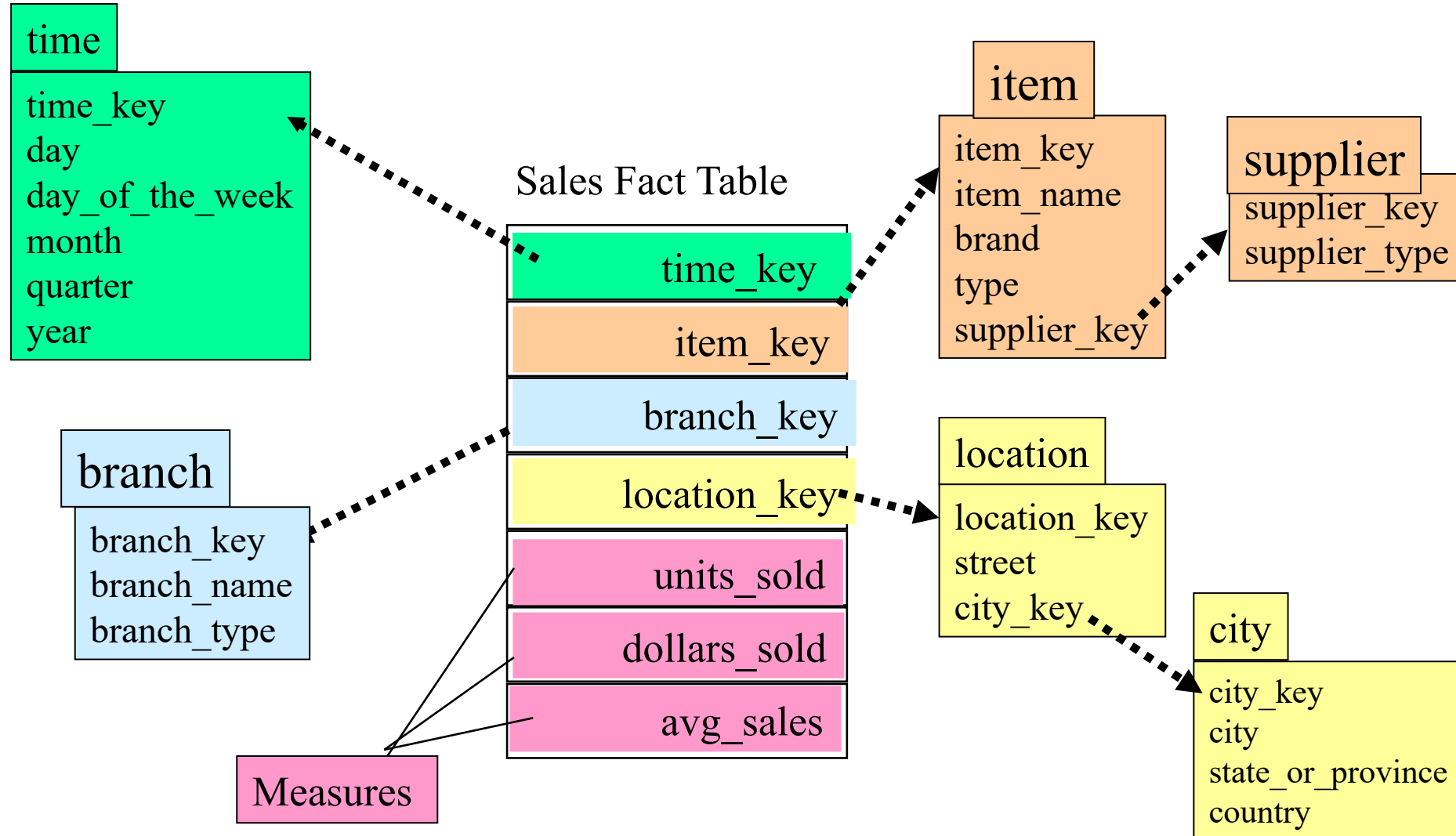
❑ Key Features of Domain Table

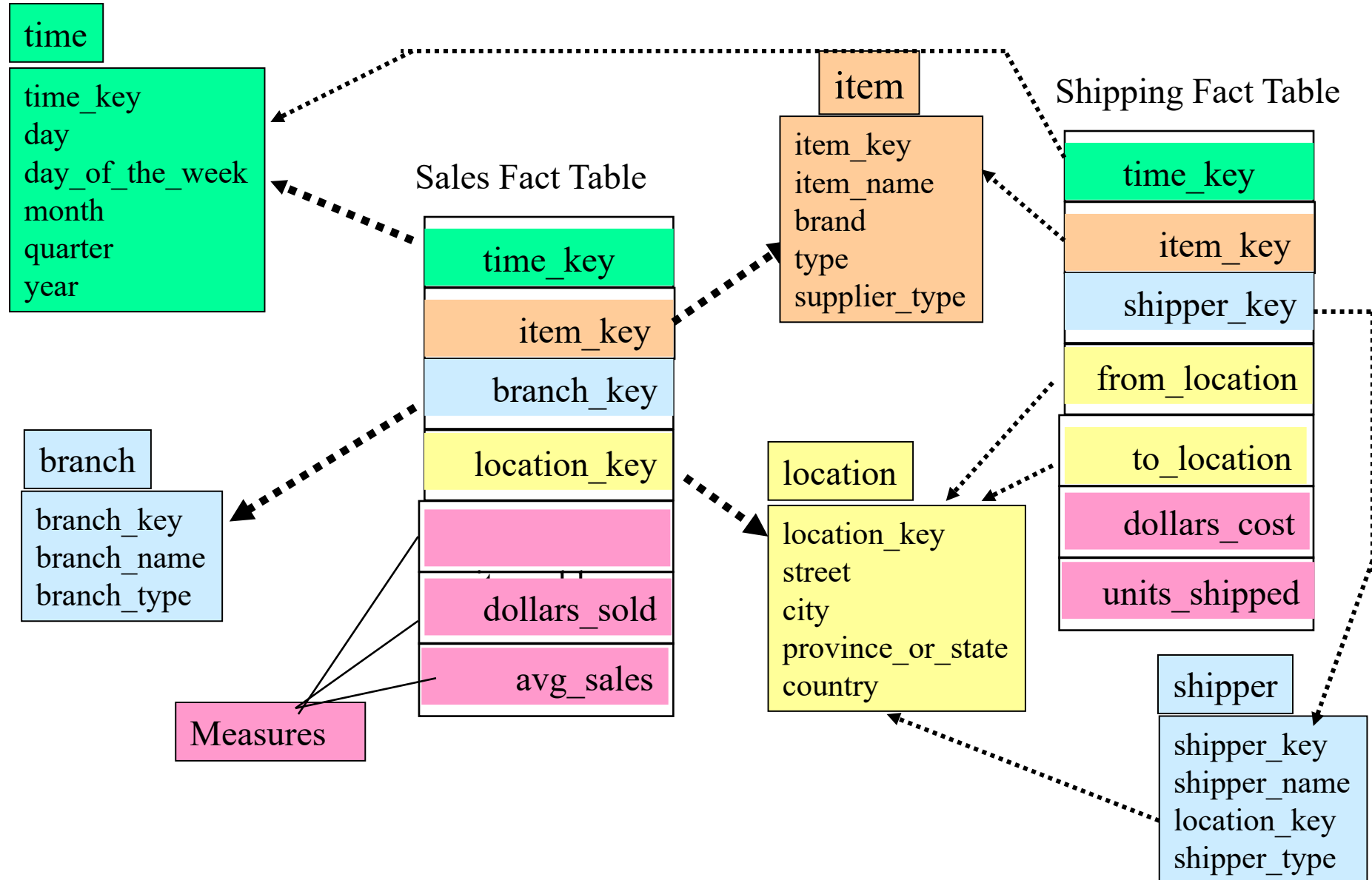
- ❑ Contains textual / descriptive data
- ❑ Smaller in size compared to fact tables
- ❑ Stores primary keys
- ❑ Rarely changes
- ❑ Used to give meaning to fact data

- ❑ Modeling data warehouses: dimensions & measures
 - ❑ Star schema: A fact table in the middle connected to a set of dimension tables
 - ❑ Snowflake schema: A refinement of star schema where some dimensional hierarchy is normalized into a set of smaller dimension tables, forming a shape similar to snowflake
 - ❑ Fact constellations: Multiple fact tables share dimension tables, viewed as a collection of stars, therefore called **galaxy schema** or fact constellation

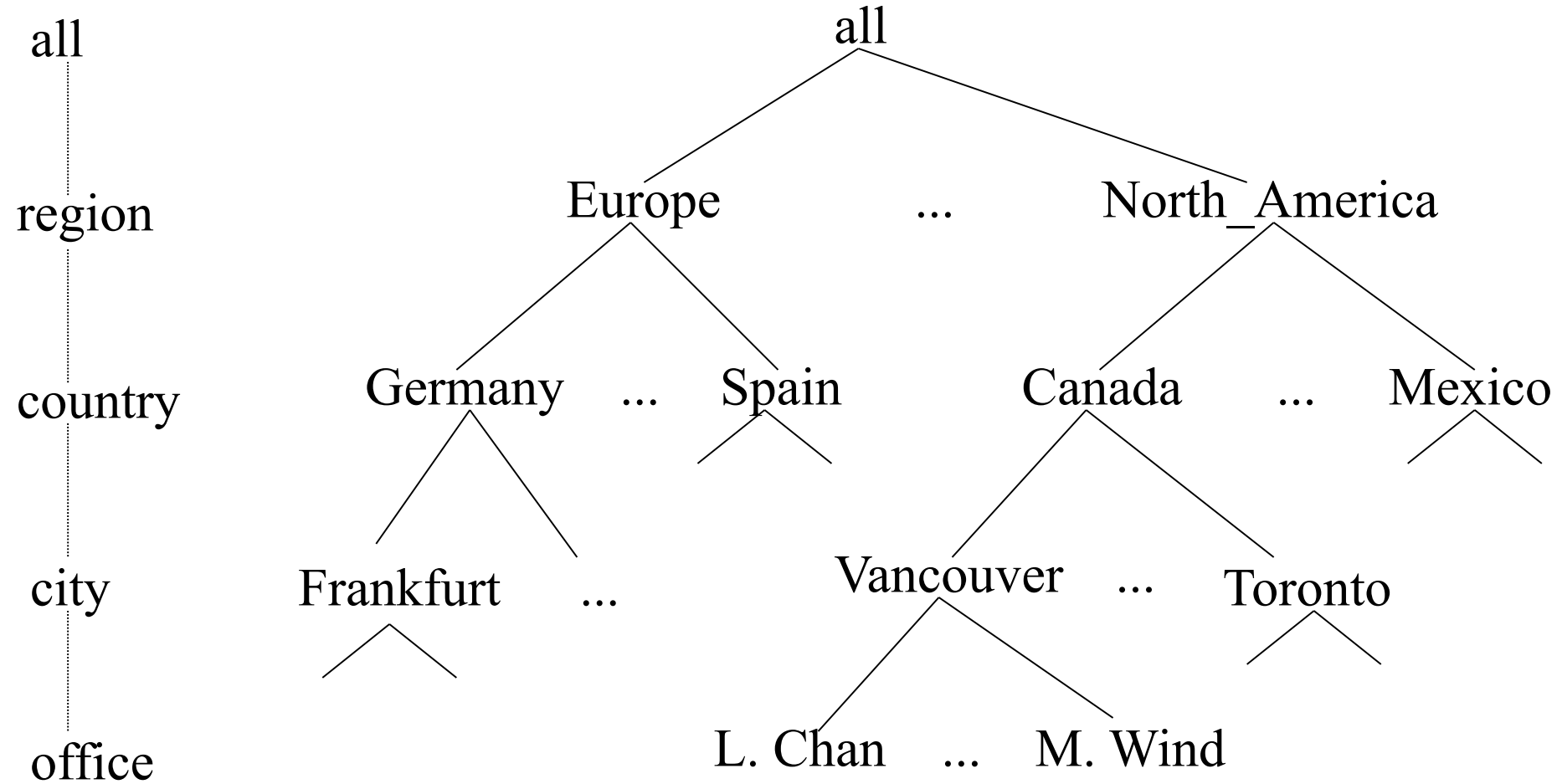
Star Schema: An Example







A Concept Hierarchy for a Dimension (location)



- ❑ A data cube measure is a numeric function that can be evaluated at each point in the data cube space.
- ❑ A measure value is computed for a given point by aggregating the data corresponding to the respective dimension–value pairs defining the given point.
- ❑ Think of a data cube as a structure where data is organized by different categories like time, product, and location.
- ❑ Each combination of these categories gives one specific point in the cube.
- ❑ Dimensions: Time = January, Product = Laptop, City = Karachi
- ❑ Now many sales records exist for this combination: 10,000, 20,000, 15,000
- ❑ We aggregate (add them): $10,000 + 20,000 + 15,000 = 45,000$
- ❑ So: Point in cube = (January, Laptop, Karachi)
 - ❑ Measure = 45,000 (total sales)

Data Cube Measures: Three Categories

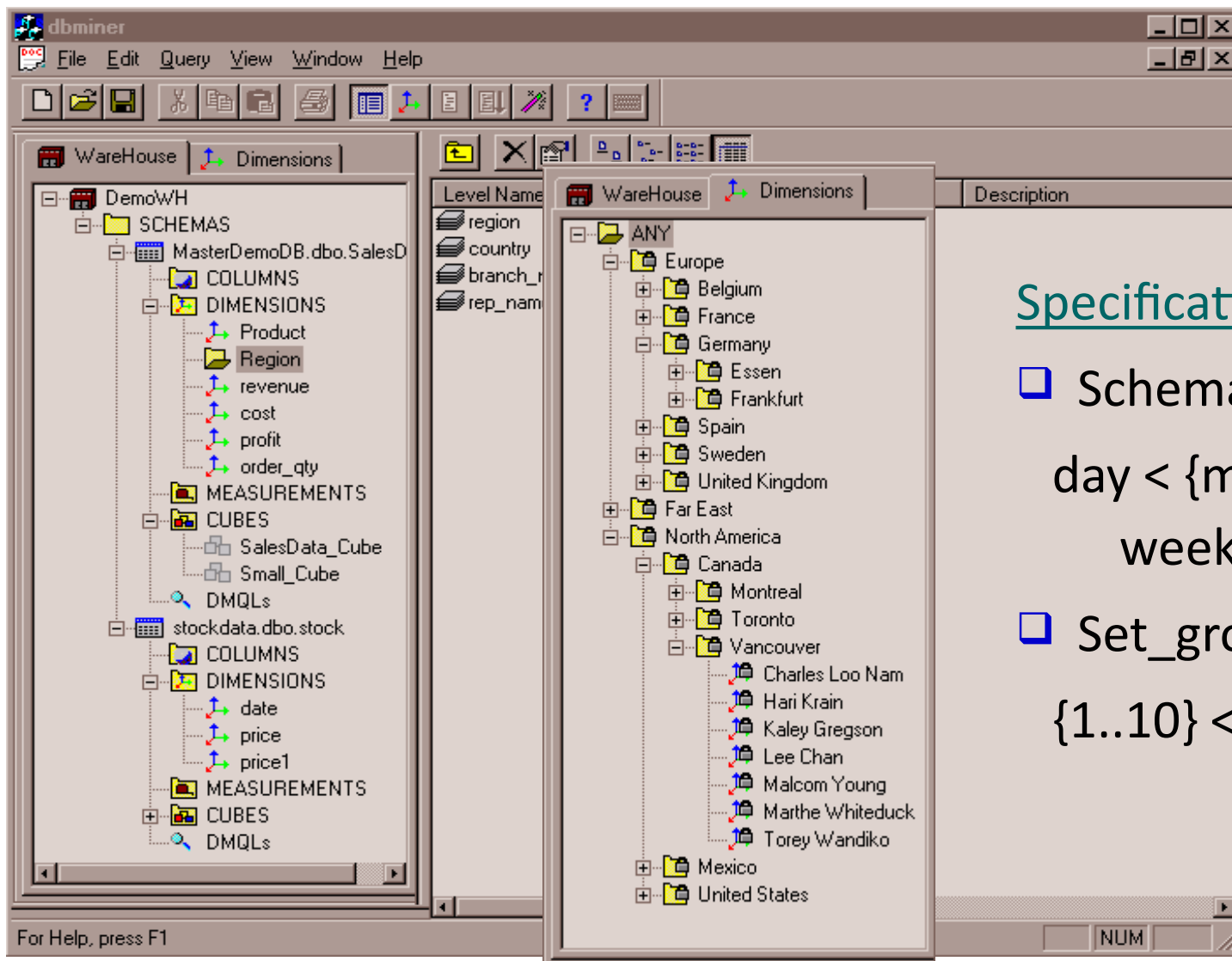
- ❑ Distributive:
- ❑ A measure is distributive if it uses a distributive function.
 - ❑ Number of items $\rightarrow$ COUNT $\rightarrow$ distributive measure
- ❑ An aggregate function is called distributive when it can be calculated in small parts first
- ❑ Then combined to get the same final result as if we had calculated everything together at once.
- ❑ Suppose total sales data is split into 3 parts: Part A = 100, Part B = 200, Part C = 300
- ❑ Now: Step 1: calculate sum in each part: A = 100, B = 200, C = 300
- ❑ Step 2: add them together: $100 + 200 + 300 = 600$
- ❑ Now compare: If we had added all data directly $\rightarrow$ also 600
 - ❑ E.g., count(), sum(), min(), max()

Data Cube Measures: Three Categories

- ❑ Algebraic:
- ❑ It is a function that can be calculated using a fixed number of simpler (distributive) functions, and then combined to get the final result.
- ❑ Instead of directly calculating a value in one step, you:
 - ❑ Break the calculation into a few small parts
 - ❑ Each part is calculated using distributive functions like SUM, COUNT, MIN, MAX
 - ❑ Then you combine those results using a formula
- ❑ If this always works with a fixed number of steps (not infinite or changing size), then the function is algebraic.
- ❑ For Example, Average = $SUM / COUNT$, Is `min_N()` an algebraic measure? How about `standard_deviation()`?
- ❑ So we calculate: SUM of all values (using distributive function), COUNT of all values (also distributive)
- ❑ Then combine: $AVG = SUM \div COUNT$
- ❑ This is why AVG is an algebraic function

Data Cube Measures: Three Categories

- ❑ Holistic:
- ❑ A measure is holistic if it is obtained by applying a holistic aggregate function.
- ❑ A holistic aggregate function is the most “complex” type of aggregate function.
- ❑ It is a function that cannot be broken into fixed small parts (like SUM or AVG) and then combined to get the final answer.
- ❑ if there is no constant bound on the storage size needed to describe a sub-aggregate.
 - ❑ E.g., median(), mode(), rank()
 - ❑ Mode = most frequent value
 - ❑ To find it, you must know: frequency of every value in the entire dataset
 - ❑ You cannot reliably merge small summaries to get the final mode without keeping lots of information.

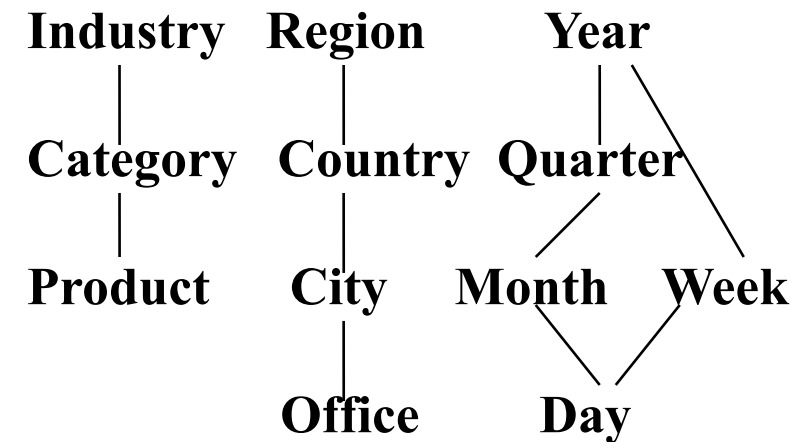
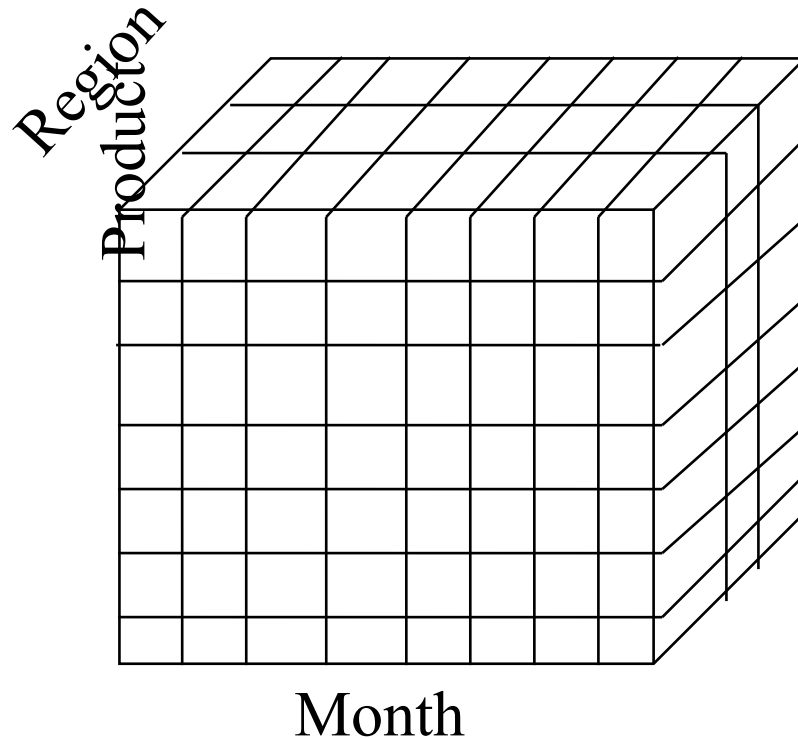


Specification of hierarchies

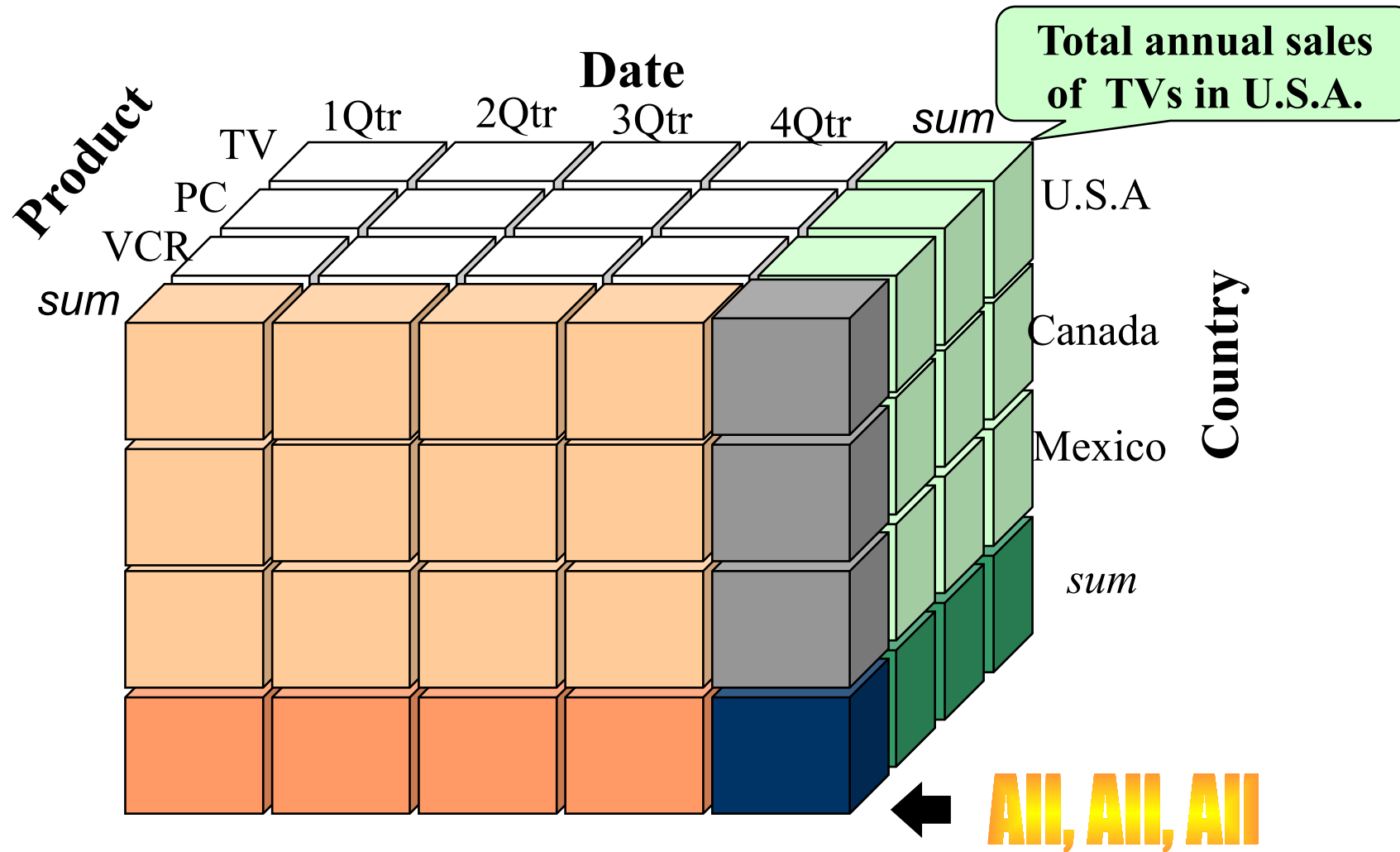
- Schema hierarchy
 - day < {month < quarter;
week} < year
- Set_grouping hierarchy
 - {1..10} < inexpensive

- ❑ Sales volume as a function of product, month, and region

Dimensions: *Product, Location, Time*
Hierarchical summarization paths

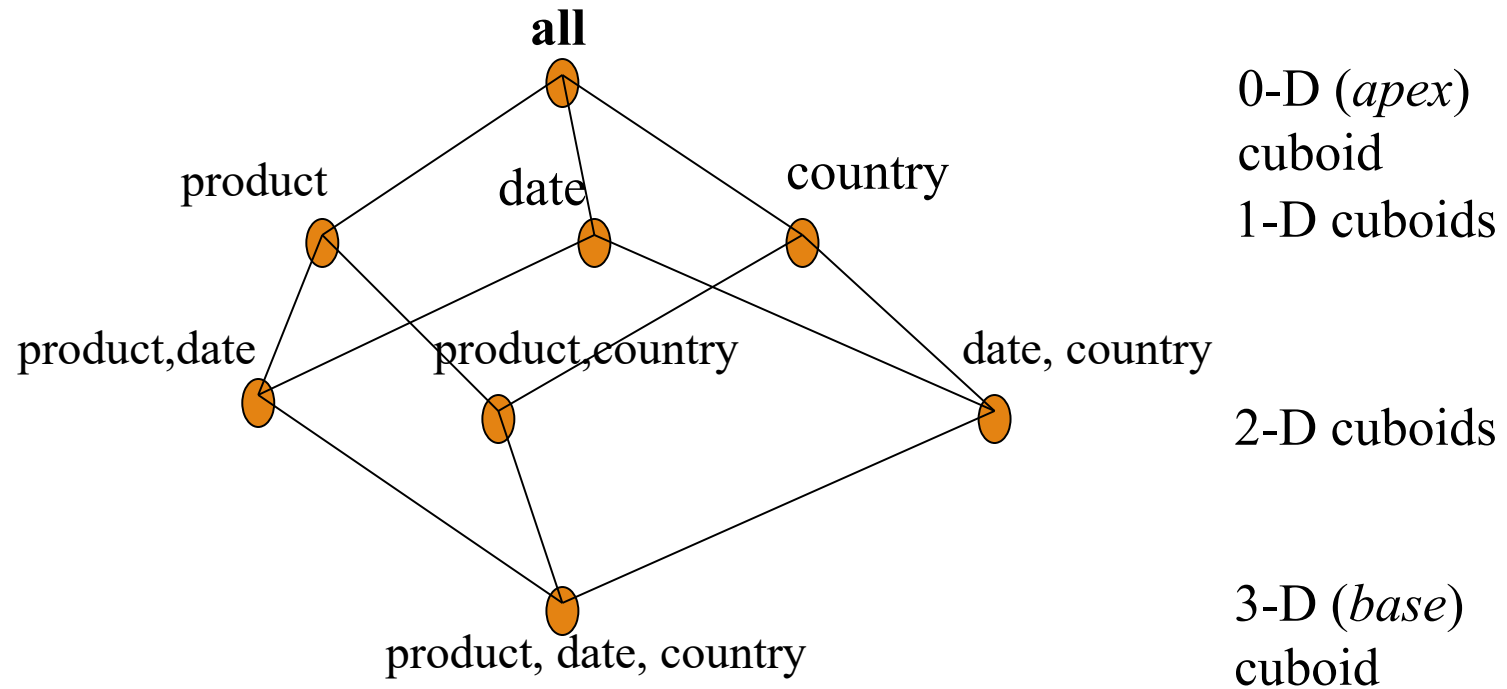


A Sample Data Cube





Cuboids Corresponding to the Cube



- ❑ **Roll up (drill-up):** summarize data
 - ❑ *by climbing up hierarchy or by dimension reduction*
 - ❑ *Combine detailed data into higher-level groups to get a summarized view*
 - ❑ Roll-up performs aggregation (like SUM, COUNT) while moving up the hierarchy.
 - ❑ Roll-up can also work by removing a dimension completely.
- ❑ **Drill down (roll down):** reverse of roll-up
 - ❑ *from higher level summary to lower level summary or detailed data, or introducing new dimensions*
 - ❑ Breaks aggregated data into smaller parts, Adds more detail (by adding a new dimension), Moves down the hierarchy
 - ❑ *It means moving from summary data to more detailed data, so you can see the finer details hidden inside the data.*
 - ❑ In OLAP, dimensions have hierarchies. For Time: Year → Quarter → Month → Day

❑ **Slice and dice:** *project and select*

- ❑ A slice operation means you fix one dimension to a single value, and then you get a smaller cube (a subcube).
 - ❑ It is like cutting the cube in one direction.
 - ❑ For example, imagine a sales cube with dimensions: Time, Location, Item
 - ❑ Now you apply a condition: Time = “Q1”
 - ❑ This means:
 - ❑ We only look at data for Quarter 1
 - ❑ All other time values are removed
- ❑ A dice operation is more complex. It selects multiple conditions across multiple dimensions, creating a smaller subcube.
- ❑ It is like cutting the cube from many sides at once.
- ❑ For example, we apply:

❑ Slice and dice: *project and select*

- ❑ A dice operation is more complex. It selects multiple conditions across multiple dimensions, creating a smaller subcube.
- ❑ It is like cutting the cube from many sides at once.
- ❑ For example, we apply:
- ❑ Location = Toronto or Vancouver, Time = Q1 or Q2, Item = Home entertainment or Computer
- ❑ Now we are filtering on three dimensions at the same time.

❑ Pivot (rotate):

- ❑ *reorient the cube, visualization, 3D to series of 2D planes*
- ❑ *way of changing the way data is viewed without changing the data itself. It helps you look at the same information from a different angle.*
- ❑ *Changes the orientation of data axes*
- ❑ *Shows the same data in a different layout*
- ❑ *Helps users see relationships more clearly*
- ❑ *In a 3D data cube (Time, Location, Product), pivot can:*
 - ❑ *Rotate the cube, Bring a different dimension to the front*
- ❑ *Convert 3D view into different 2D views*
- ❑ *So instead of always looking at:*
- ❑ *Time vs Location*
- ❑ *You can pivot to: Product vs Time, Product vs Location*

Suppose you are viewing a 2D table:

City \ Product	Laptop	Phone
Karachi	50000	30000
Lahore	40000	25000

Now you apply pivot, and rotate the axes:

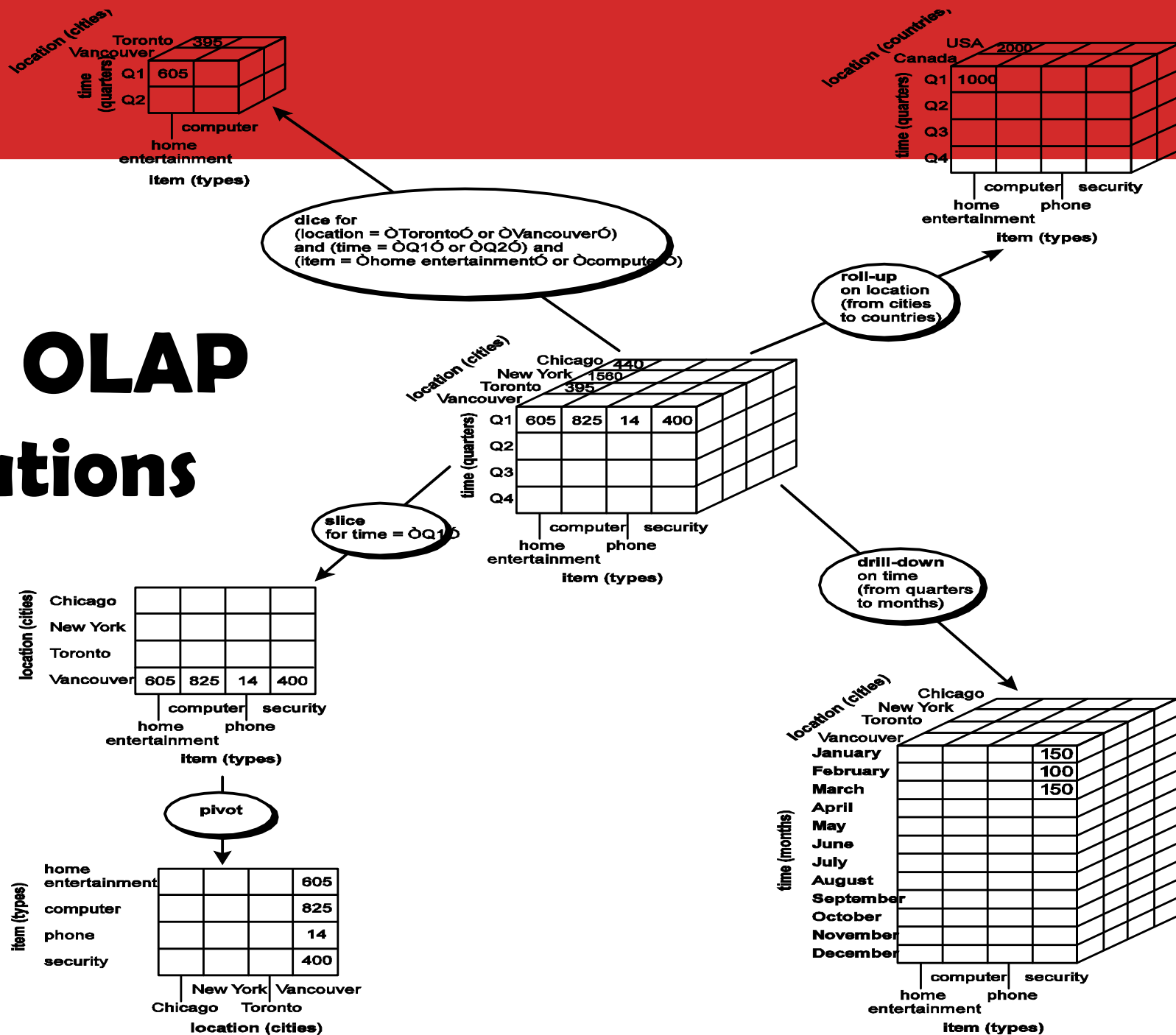
Product \ City	Karachi	Lahore
Laptop	50000	40000
Phone	30000	25000

- ❑ Other operations
 - ❑ *Drill across: involving (across) more than one fact table*
 - ❑ sometimes a business needs to combine multiple fact tables, such as:
 - ❑ Sales data (how much sold)
 - ❑ Shipping data (delivery information)
 - ❑ So instead of looking at only sales, you might compare:
 - ❑ Sales vs Shipping time
 - ❑ Profit vs Return rates

- **Drill through:** *through the bottom level of the cube to its back-end relational tables (using SQL)*
 - *Going from summary to raw detailed records in database*
 - Drill-through means going from the cube down to the original database tables (SQL level).
 - For example:
 - Cube shows: “Total sales = 50,000 in Karachi”
 - Drill-through shows: actual customer transactions behind that number
- Other OLAP operations may include ranking the top N or bottom N items in lists, as well as computing moving averages, growth rates, interests, internal return rates, depreciation, currency conversions, and statistical functions



Typical OLAP Operations



- ❑ A statistical database is a database system that is designed to support statistical applications
- ❑ OLAP (Online Analytical Processing) → used in business (sales, marketing, finance)
- ❑ Statistical Databases (SDBs) → used in statistics and research (population, economy, health data)
- ❑ Both systems allow you to analyze data from different levels (summary and detail).

❑ Application area

- ❑ OLAP → business problems

- ❑ (sales, customers, profits, marketing)

- ❑ SDB → socioeconomic and research data

- ❑ (population, income, education, health)

- ❑ So OLAP is commercial, SDB is statistical/research-based.

❑ Privacy concern

- ❑ SDB systems deal with sensitive data like:

 - ❑ income of people, population details, health records

- ❑ So privacy is a big issue.

❑ For example:

- ❑ You may see average income of a city

- ❑ But accessing individual person's income may be restricted

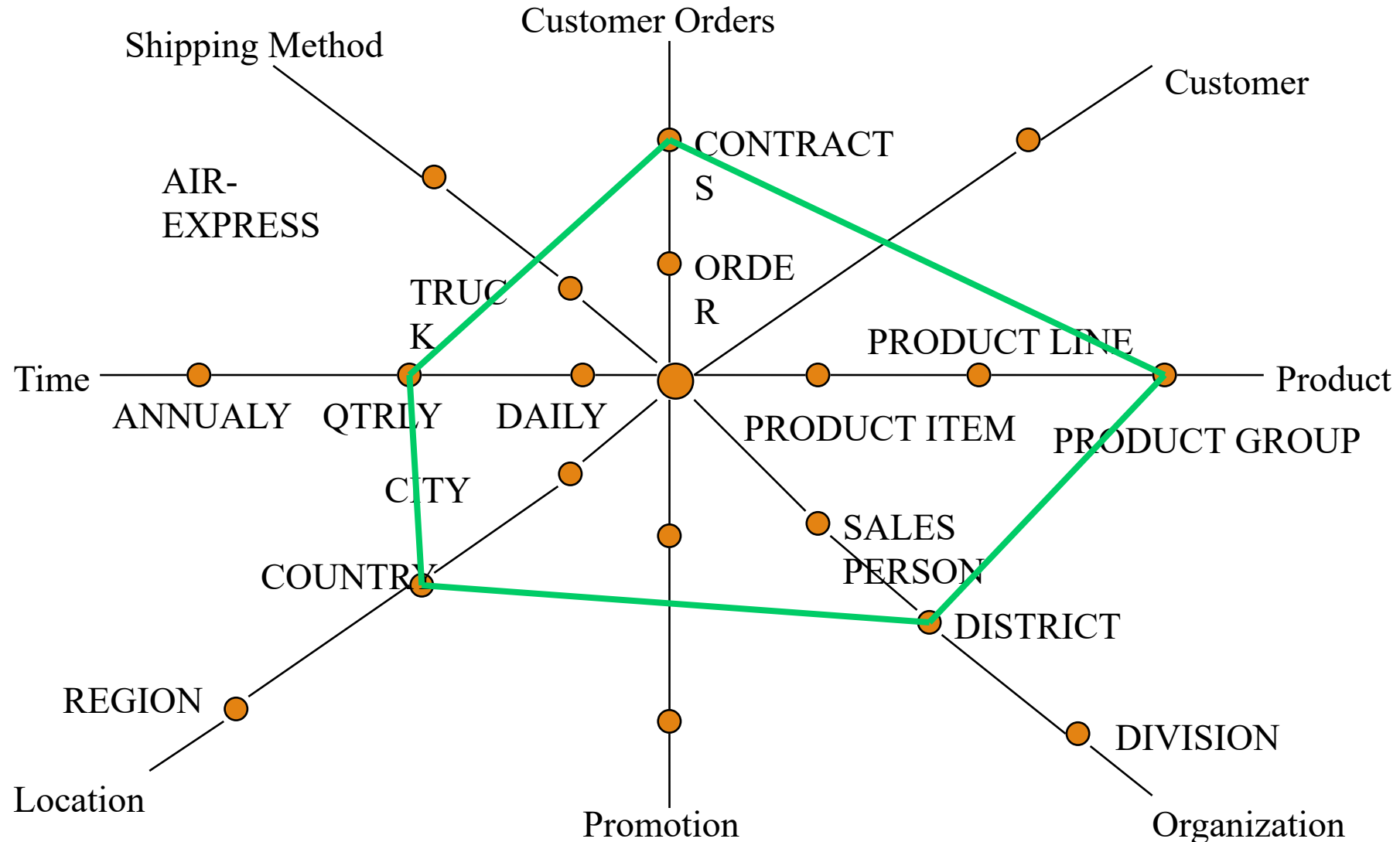
- ❑ So SDB focuses heavily on protecting low-level data

❑ Data volume handling

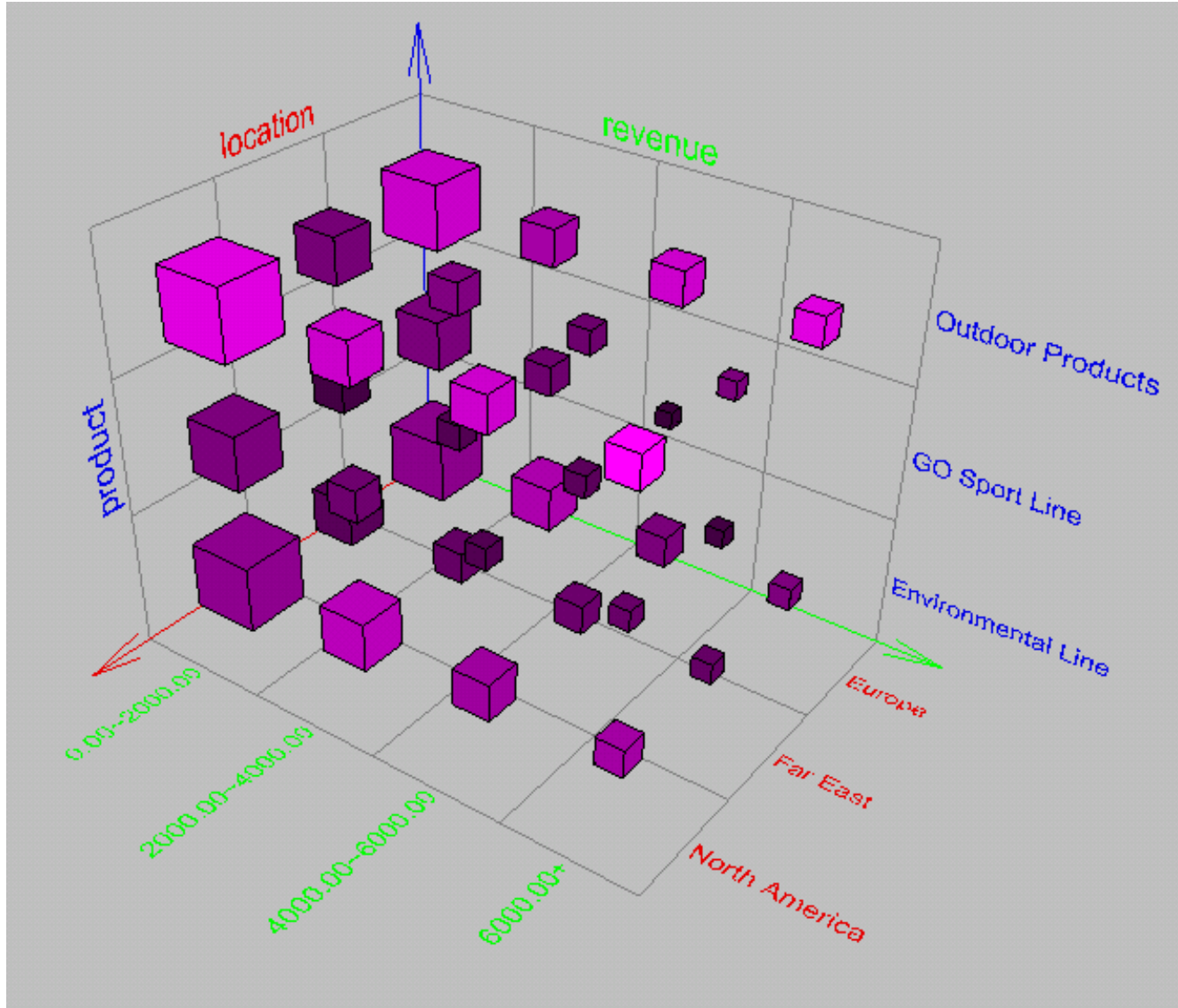
- ❑ OLAP → designed to handle huge business data efficiently
- ❑ SDB → also handles large data, but not optimized like OLAP for fast business queries
- ❑ OLAP systems are more performance-optimized for large-scale querying.

- ❑ A starnet model is just a way to visually organize and understand a multidimensional database (like an OLAP system) in a simple structure.
- ❑ Imagine you place a main point in the center, and from that point, you draw lines outward in different directions.
- ❑ Each line represents a dimension (like Time, Location, Product).
- ❑ So it looks like a star with rays coming out, which is why it is called a starnet model.
- ❑ Each dimension is shown as a line
- ❑ Each line has levels of detail called footprints

A Star-Net Query Model



Browsing a Data Cube



- ☐ Visualization
- ☐ OLAP capabilities
- ☐ Interactive manipulation

- ❑ Four views regarding the design of a data warehouse
 - ❑ Top-down view
 - ❑ allows selection of the relevant information necessary for the data warehouse
 - ❑ Data source view
 - ❑ exposes the information being captured, stored, and managed by operational systems
 - ❑ Data warehouse view
 - ❑ consists of fact tables and dimension tables
 - ❑ Business query view
 - ❑ sees the perspectives of data in the warehouse from the view of end-user

- ❑ **Top-down, bottom-up approaches or a combination** of both
 - ❑ Top-down: Starts with overall design and planning (mature)
 - ❑ Bottom-up: Starts with experiments and prototypes (rapid)
- ❑ **From software engineering point of view**
 - ❑ Waterfall: structured and systematic analysis at each step before proceeding to the next
 - ❑ Spiral: rapid generation of increasingly functional systems, short turn around time, quick turn around
- ❑ **Typical data warehouse design process**
 - ❑ Choose a business process to model, e.g., orders, invoices, etc.
 - ❑ Choose the grain (*atomic level of data*) of the business process
 - ❑ Choose the dimensions that will apply to each fact table record
 - ❑ Choose the measure that will populate each fact table record

- ❑ Three kinds of data warehouse applications
 - ❑ Information processing
 - ❑ supports querying, basic statistical analysis, and reporting using crosstabs, tables, charts and graphs
 - ❑ Analytical processing
 - ❑ multidimensional analysis of data warehouse data
 - ❑ supports basic OLAP operations, slice-dice, drilling, pivoting
 - ❑ Data mining
 - ❑ knowledge discovery from hidden patterns
 - ❑ supports associations, constructing analytical models, performing classification and prediction, and presenting the mining results using visualization tools

❑ Why **online analytical mining**?

- ❑ High quality of data in data warehouses
 - ❑ DW contains integrated, consistent, cleaned data
- ❑ Available information processing structure surrounding data warehouses
 - ❑ ODBC(Open Database Connectivity), OLEDB(Object Linking and Embedding Database), Web accessing, service facilities, reporting and OLAP tools
- ❑ OLAP-based exploratory data analysis
 - ❑ Mining with drilling, dicing, pivoting, etc.
- ❑ On-line selection of data mining functions
 - ❑ Integration and swapping of multiple mining functions, algorithms, and tasks

□ Data cube can be viewed as a lattice of cuboids

□ The bottom-most cuboid is the base cuboid

□ The top-most cuboid (apex) contains only one cell

□ How many cuboids in an n-dimensional cube with L levels?

Why this formula?

$$T = \prod_{i=1}^n (L_i + 1)$$

□ Materialization of data cube

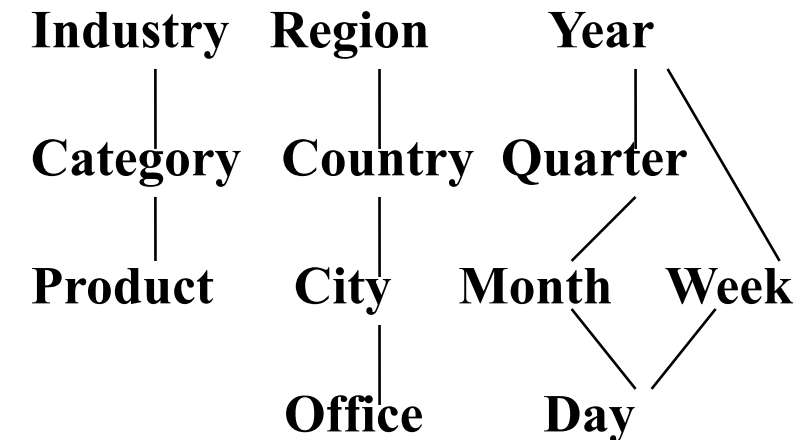
□ **Full materialization:** Materialize every (cuboid)

□ **No materialization:** Materialize none (cuboid)

□ **Partial materialization:** Materialize some cuboids

□ Which cuboids to materialize?

□ Selection based on size, sharing, access frequency, etc.



- ❑ Cube definition and computation in DMQL

```
define cube sales [item, city, year]: sum (sales_in_dollars)
```

```
compute cube sales
```

- ❑ Transform it into a SQL-like language (with a new operator **cube by**, introduced by Gray et al.'96)

```
SELECT item, city, year, SUM (amount)
```

```
FROM SALES
```

```
CUBE BY item, city, year
```

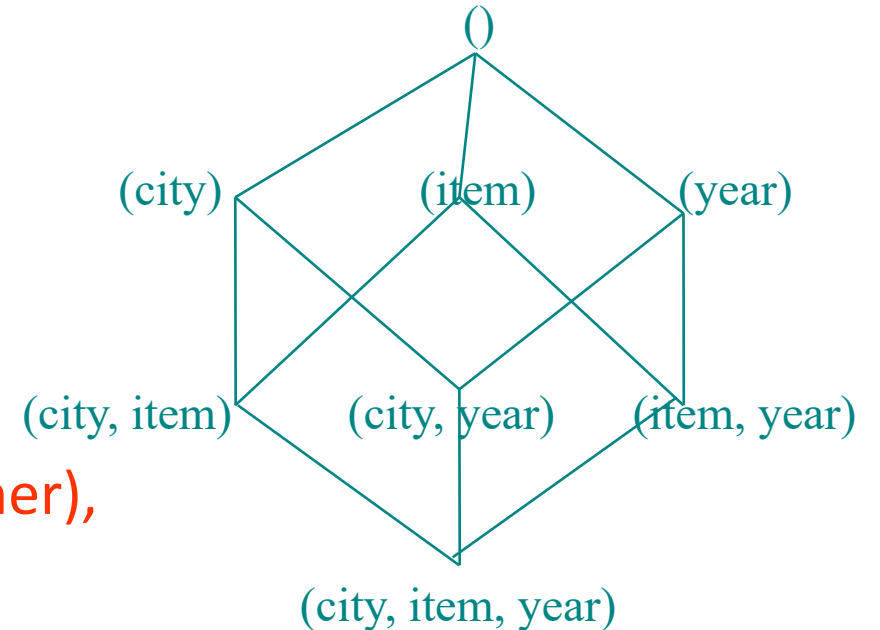
- ❑ Need compute the following Group-Bys

```
(date, product, customer),
```

```
(date, product),(date, customer), (product, customer),
```

```
(date), (product), (customer)
```

```
()
```



- Index on a particular column
 - Each value in the column has a bit vector: bit-op is fast
 - The length of the bit vector: # of records in the base table
 - The i -th bit is set if the i -th row of the base table has the value for the indexed column
 - not suitable for high cardinality domains
- A recent bit compression technique, Word-Aligned Hybrid (WAH), makes it work for high cardinality domain as well [Wu, et al. TODS'06]

Base table

Cust	Region	Type
C1	Asia	Retail
C2	Europe	Dealer
C3	Asia	Dealer
C4	America	Retail
C5	Europe	Dealer

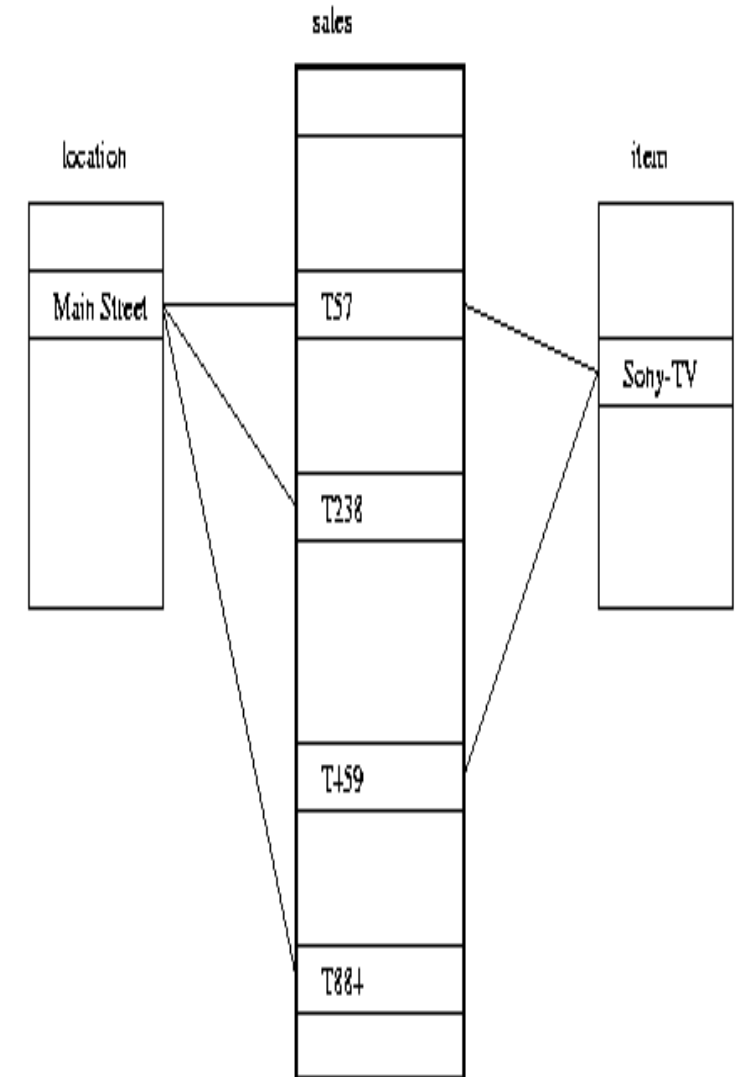
Index on Region

RecID	Asia	Europe	America
1	1	0	0
2	0	1	0
3	1	0	0
4	0	0	1
5	0	1	0

Index on Type

RecID	Retail	Dealer
1	1	0
2	0	1
3	0	1
4	1	0
5	0	1

- ❑ Join index: $JI(R\text{-id}, S\text{-id})$ where $R(R\text{-id}, \dots) \triangleright \triangleleft S(S\text{-id}, \dots)$
- ❑ Traditional indices map the values to a list of record ids
 - ❑ It materializes relational join in JI file and speeds up relational join
- ❑ In data warehouses, join index relates the values of the dimensions of a star schema to rows in the fact table.
 - ❑ E.g., fact table: *Sales* and two dimensions *city* and *product*
 - ❑ A join index on *city* maintains for each distinct city a list of R-IDs of the tuples recording the Sales in the city
 - ❑ Join indices can span multiple dimensions



- ❑ **Determine which operations** should be performed on the available cuboids
 - ❑ Transform drill, roll, etc. into corresponding SQL and/or OLAP operations, e.g., dice = selection + projection
- ❑ **Determine which materialized cuboid(s)** should be selected for OLAP op.
 - ❑ Let the query to be processed be on $\{brand, province_or_state\}$ with the condition “ $year = 2004$ ”, and there are 4 materialized cuboids available:
 - 1) $\{year, item_name, city\}$
 - 2) $\{year, brand, country\}$
 - 3) $\{year, brand, province_or_state\}$
 - 4) $\{item_name, province_or_state\}$ where $year = 2004$Which should be selected to process the query?
- ❑ Explore indexing structures and compressed vs. dense array structs in MOLAP

❑ Relational OLAP (ROLAP)

- ❑ Use relational or extended-relational DBMS to store and manage warehouse data and OLAP middle ware
- ❑ Include optimization of DBMS backend, implementation of aggregation navigation logic, and additional tools and services
- ❑ Greater scalability

❑ Multidimensional OLAP (MOLAP)

- ❑ Sparse array-based multidimensional storage engine
- ❑ Fast indexing to pre-computed summarized data

❑ Hybrid OLAP (HOLAP) (e.g., Microsoft SQLServer)

- ❑ Flexibility, e.g., low level: relational, high-level: array

❑ Specialized SQL servers (e.g., Redbricks)

- ❑ Specialized support for SQL queries over star/snowflake schemas

- ❑ Data warehousing: A multi-dimensional model of a data warehouse
 - ❑ A data cube consists of *dimensions & measures*
 - ❑ Star schema, snowflake schema, fact constellations
 - ❑ OLAP operations: drilling, rolling, slicing, dicing and pivoting
- ❑ Data Warehouse Architecture, Design, and Usage
 - ❑ Multi-tiered architecture
 - ❑ Business analysis design framework
 - ❑ Information processing, analytical processing, data mining, OLAM
- ❑ Implementation: Efficient computation of data cubes
 - ❑ Partial vs. full vs. no materialization
 - ❑ Indexing OALP data: Bitmap index and join index
 - ❑ OLAP query processing
 - ❑ OLAP servers: ROLAP, MOLAP, HOLAP

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